

TASK 1

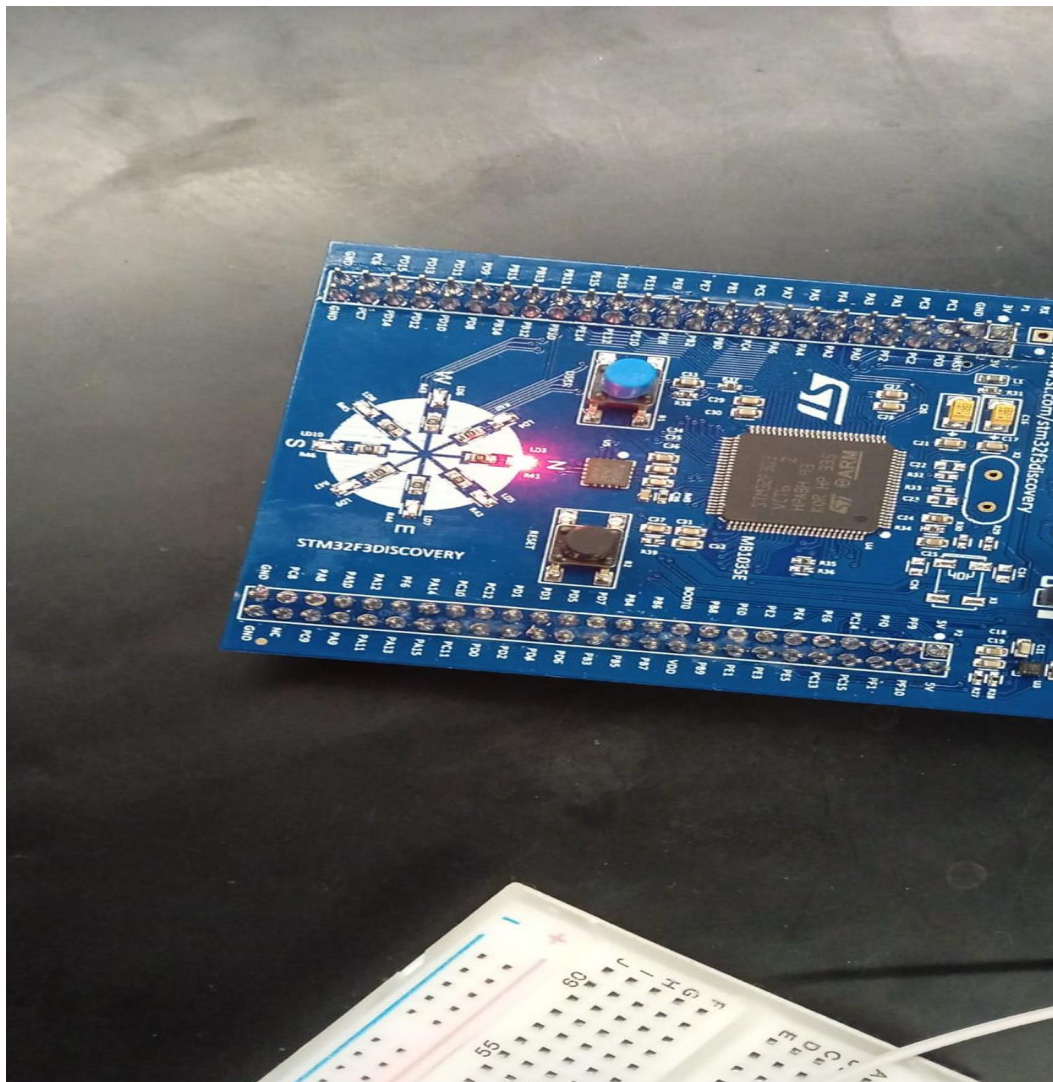
```
void delay_ms(uint32_t ms){
    __HAL_TIM_SET_COUNTER(&htim2, 0);
    HAL_TIM_Base_Start(&htim2);

    while (__HAL_TIM_GET_COUNTER(&htim2) < ms);

    HAL_TIM_Base_Stop(&htim2);
}

while (1)
{
    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_9, GPIO_PIN_SET); // RED LED ON
    delay_ms(1000);

    HAL_GPIO_WritePin(GPIOE, GPIO_PIN_9, GPIO_PIN_RESET); // RED LED OFF
    delay_ms(1000);
}
```



TASK 2

CODE:

```
//TASK 2
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
    if (htim->Instance == TIM2)
    {
        HAL_GPIO_TogglePin(GPIOE, GPIO_PIN_9); // Red (North) LED
    }
}

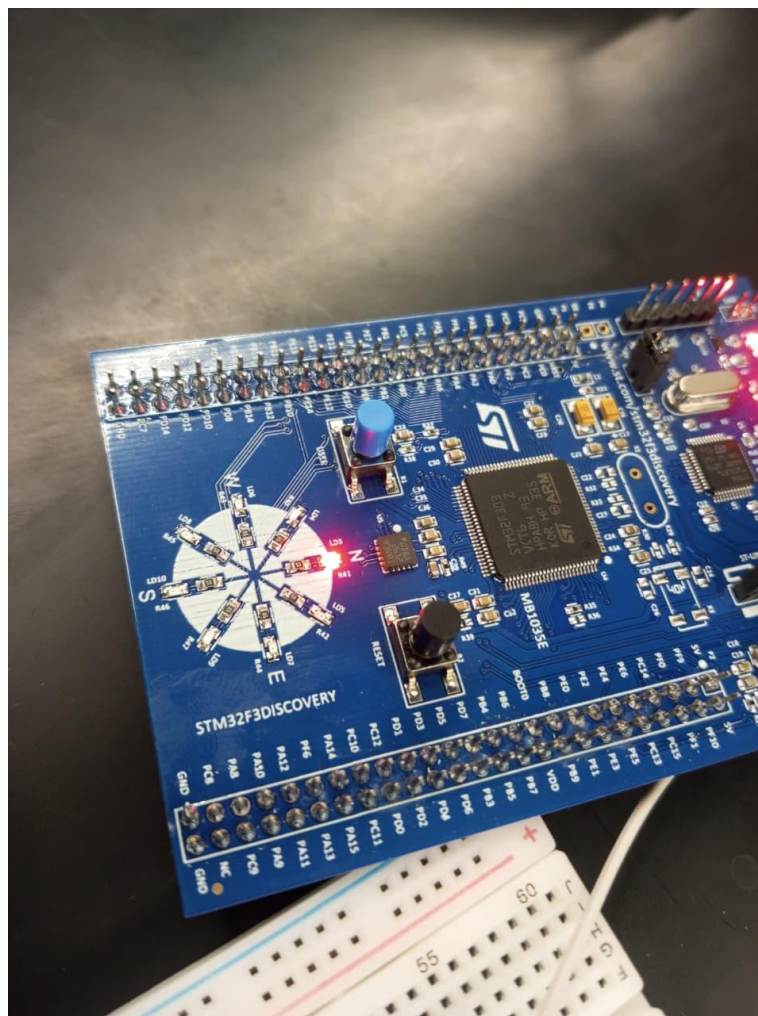
HAL_TIM_Base_Start_IT(&htim2);

/* USER CODE END 2 */

/* USER CODE BEGIN WHILE */
while (1)
{
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
}
```

OUTPUT:



TASK 3:

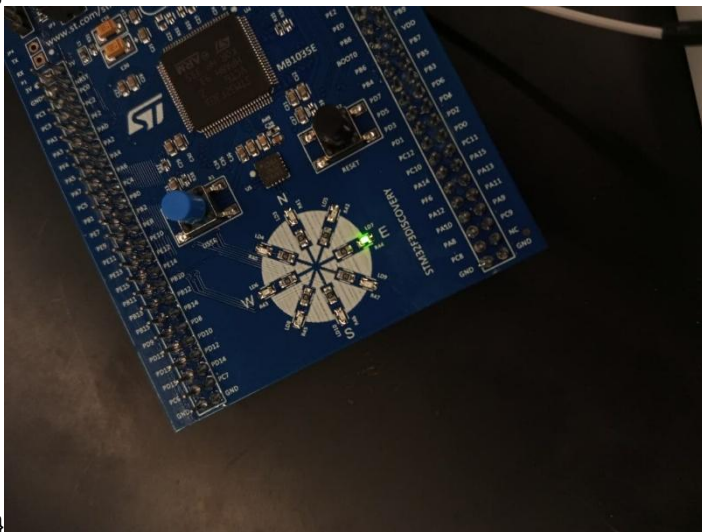
CODE:

```
/* ----- TASK 3----- */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
    if(htim->Instance == TIM2)
    {
        // Increment software counters
        countA++;
        countB++;
        countC++;

        // LED1: PE9 (Red)
        if(countA >= THRESHOLD_A)
        {
            HAL_GPIO_TogglePin(GPIOE, GPIO_PIN_9);
            countA = 0;
        }

        // LED2: PE10 (Green)
        if(countB >= THRESHOLD_B)
        {
            HAL_GPIO_TogglePin(GPIOE, GPIO_PIN_10);
            countB = 0;
        }

        // LED3: PE11 (Blue)
        if(countC >= THRESHOLD_C)
        {
            HAL_GPIO_TogglePin(GPIOE, GPIO_PIN_11);
            countC = 0;
        }
    }
}
```



TASK 4:

CODE:

```
void HAL_TIM_IC_CaptureCallback(TIM_HandleTypeDef *htim)
{
    if(htim->Instance == TIM3)
    {
        if(htim->Channel == HAL_TIM_ACTIVE_CHANNEL_1)
        {
            // Read the captured value
            uint32_t current_capture = HAL_TIM_ReadCapturedValue(htim, TIM_CHANNEL_1);
            // Calculate period (handle overflow)
            if(current_capture >= last_capture)
            {
                period = current_capture - last_capture;
            }
            else
            {
                // Timer overflow occurred
                period = (0xFFFF - last_capture) + current_capture + 1;
            }
            // Update last capture value
            last_capture = current_capture;
            // Calculate frequency
            // Frequency = Timer_Clock / [(Prescaler + 1) × Period]
            if(period > 0)
            {
                frequency = (float)TIMER_CLOCK_FREQ / ((float)(TIM3_PRESCALER + 1) * (float)period);
                valid_measurement = 1;
            }
            // Reset overflow counter
            overflow_count = 0;
        }
    }
}
```

[illegible][illegible]

Calculations:

1. Timer Clock Configuration

Base Clock Calculation:

$$\begin{aligned}\text{System Clock} &= 8 \text{ MHz} \\ \text{PLL multiplier} &= 6 \\ \text{System Clock after PLL} &= 8 \text{ MHz} \times 6 = 48 \text{ MHz} \\ \text{APB1 Clock} &= 24 \text{ MHz}\end{aligned}$$

Timer Clock Frequency:

APB1 Timer Clock = 48 MHz

TIMER_CLOCK_FREQ = 48,000,000 Hz

2. TIM3 Configuration

From the code:

Prescaler = 1;

Effective Timer Clock:

$$\begin{aligned}\text{Effective Timer Frequency} &= \text{CLOCK_FREQ} / (\text{Prescaler} + 1) \\ &= 48,000,000 / (1 + 1) \\ &= 48,000,000 / 2 \\ &= 24 \text{ MHz} = 24,000,000 \text{ Hz}\end{aligned}$$

Timer Resolution:

$$\begin{aligned}\text{Timer Tick Period} &= 1 / 24,000,000 \text{ Hz} \\ &= 0.04167 \mu\text{s} \\ &= 41.67 \text{ ns}\end{aligned}$$

3. Frequency Calculation Formula

From the code:

$$\text{frequency} = \text{CLOCK_FREQ} / (\text{PRESCALER} + 1) * \text{period};$$

Simplified:

$$\begin{aligned}\text{Frequency} &= 48,000,000 / (2 \times \text{period}) \\ &= 24,000,000 / \text{period Hz}\end{aligned}$$

4. Example Calculations

Example 1: 1 kHz Input Signal

Input Signal Period = $1 / 1,000 \text{ Hz} = 1 \text{ ms} = 0.001 \text{ s}$

$$\begin{aligned}\text{Number of Timer Ticks (period)} &= 0.001 \text{ s} / (1 / 24,000,000) \\ &= 0.001 \times 24,000,000 \\ &= 24,000 \text{ ticks}\end{aligned}$$

$$\begin{aligned}\text{Measured Frequency} &= 24,000,000 / 24,000 \\ &= 1,000 \text{ Hz} = 1 \text{ kHz}\end{aligned}$$